Problem 00000000567: the ordinary representation category already fails in type A_1

lizaixi01 – 24 October 2026

Statement and scope

The first clause asserts that the number of pairwise nonisomorphic maximal Ext-orthogonal objects in the idempotent completion of the representation category of a Dynkin quiver is

$$\prod_{i=1}^{n} \frac{e_i + h + 1}{e_i + 1}.$$

Take the ordinary category of finite-dimensional complex representations of the one-vertex, no-arrow quiver A_1. Let $\mathcal{E}$ be its Karoubi envelope. We prove two incompatible counts: with all object isomorphism classes there are infinitely many; with the customary additional *basic* convention there is exactly one. The displayed product is 2.

We use the standard additive meaning of maximal rigidity, rather than a strict ordering of objects that would forbid repeated summands. Write $X \in \operatorname{add} T$ when X is a direct summand of a finite direct sum of copies of T. An object T is maximal rigid if

$$\operatorname{Ext}^1(T,T) = 0, \qquad (\operatorname{Ext}^1(U,U) = 0,\ T \in \operatorname{add} U) \Longrightarrow U \in \operatorname{add} T.$$

We also prove the stronger, two-sided cluster-tilting condition

$$\operatorname{add} T = \{X : \operatorname{Ext}^1(T,X) = 0\} = \{X : \operatorname{Ext}^1(X,T) = 0\}.$$

These are the add-closure conditions in Burban–Iyama–Keller–Reiten, Definition 1.1 [1]. Basic objects are finite direct sums of pairwise nonisomorphic nonzero indecomposables.

This disproof concerns the printed *ordinary representation category*. It makes no assertion against the usual Catalan enumeration in the derived orbit cluster category $D^b(\operatorname{mod} H)/(\tau^{-1}[1])$, which is a different category [2]. The preprojective-algebra clause is not used.

The complete category and its idempotents

A representation of this quiver consists of one finite-dimensional vector space: there are no arrow maps. Evaluation at the vertex and the constant representation functor are inverse equivalences. In Lean the category is literally the functor category from the native path category of the actual arrowless quiver to `FGModuleCat` over $\mathbb{C}$.

For an arbitrary Karoubi object (V,p) with $p^2 = p$, evaluation yields an actual linear idempotent on V. Its range is finite dimensional. The maps

$$V \xrightarrow{p} \operatorname{im} p \xrightarrow{\iota} V$$

satisfy $\iota p = p$ on V and $p\iota = 1$ on $\operatorname{im} p$ (composition is written in function order here). Their representation-level lifts give an explicit isomorphism of (V,p) with the identity idempotent on $\operatorname{im} p$. Thus all objects of $\mathcal{E}$, including arbitrary formal direct factors, are covered by the genuine equivalence $\mathcal{E} \simeq \operatorname{Vect}_{\mathbb{C},\mathrm{fd}}$.

Actual Ext, finite sums, and classification

Every epimorphism between finite-dimensional vector spaces is surjective and admits the required linear lifts. Hence every object is projective. Equivalence transports projectivity to every object of $\mathcal{E}$, and the native abelian-category groups satisfy

$$\mathrm{Ext}^1_{\mathcal{E}}(X, Y) = 0 \quad \text{for all } X, Y.$$

These are actual derived-category Ext groups, not values assigned to a replacement relation.

Let $d(X)$ be the actual finite dimension of the corresponding vector space. A basis gives $X \simeq \mathbb{C}^{d(X)}$. Native finite biproducts satisfy

$$d\left(\bigoplus_i X_i\right) = \sum_i d(X_i).$$

If $a \leq b$, extension of coordinates by zero and restriction to the first a coordinates give a genuine retraction $\mathbb{C}^a \longrightarrow \mathbb{C}^b \longrightarrow \mathbb{C}^a$. Conversely any retraction induces an injective linear map, so it cannot decrease the available target dimension.

If $T \neq 0$ and $m = d(X)$, then $m \leq m\, d(T)$. Applying the preceding retraction to a finite biproduct of m copies of T gives $X \in \operatorname{add} T$. If $T = 0$, no such finite biproduct has the nonzero object $\mathbb{C}$ as a direct summand. Consequently

$$T \text{ is maximal rigid} \iff T \neq 0 \iff T \text{ satisfies both Ext-orthogonal equalities.}$$

In particular every actual $\mathbb{C}^{m+1}$ is a cluster-tilting object under these conditions. Different dimensions prohibit isomorphisms. Taking the quotient of the *full* class of qualifying Karoubi objects by actual categorical isomorphism therefore admits an injection from $\mathbb{N}$ and is infinite. No finite-cardinality convention on an infinite type is used.

An object is indecomposable precisely when its dimension is one. Indeed a positive dimension larger than one splits as $1+(d-1)$; a dimension-one biproduct decomposition forces one summand to have dimension zero. Therefore all nonzero indecomposables are isomorphic to $\mathbb{C}$. A finite pairwise-nonisomorphic indecomposable decomposition has at most one summand. Conversely the zero object and every dimension-one object have such a decomposition. Thus

$$T \text{ basic} \iff d(T) \leq 1, \qquad T \text{ basic and maximal rigid} \iff d(T) = 1.$$

The quotient of all basic maximal-rigid objects is nonempty and a subsingleton; its actual finite cardinality is exactly 1.

Coxeter data derived from the root system

The formalization constructs the native rank-one root pairing on the complex line, with roots $1, -1$, coroot functionals $x \mapsto 2x, -2x$, and evaluation pairing. Both root families span. The singleton positive simple root is an actual native base, with simple pairing 2; its index set is explicitly identified with the quiver vertex set. This is the complexified standard A_1 root system.

The Coxeter element is the product of its enumerated simple reflections. Here the genuine reflection formula gives $s(x) = x - 2x = -x$. Its square is the identity and it is not the identity, so the actual group-theoretic order is $h = 2$. The scalar -1 is a primitive h-th root of unity. The full complex line is the -1 eigenspace, and there are no other eigenvalues. Filtering the integers $0 \leq e < h$ by the actual eigenvalue condition $(-1)^e$ gives exactly $\{1\}$, with cardinality equal to the actual rank. Thus $e_1 = 1$ is derived from the reflection representation. The rank-one data also agree with Fomin–Reading, Section 13, printed page 29 [3]. The conjectured expression is therefore the genuine product

$$\prod_{e \in \{1\}} \frac{h + e + 1}{e + 1} = \frac{2 + 1 + 1}{1 + 1} = 2.$$

It agrees with neither the infinite full-object count nor the basic count 1. This refutes the first necessary clause and hence the printed conjunction.

Formalization and reproducibility

The complete argument is in `lean/Main.lean`, namespace `TLMC567`. The table locates its main result and principal bridges.

Bridge	Declaration
Main result	`complete_counterexample`
All representations and their envelope	`envelopeEquivalence`
Arbitrary actual idempotent image	`explicitImageIso`
Native Ext vanishing	`envelope_ext_vanishes`
Standard maximality characterization	`maximalRigid_iff_nonzero`
Two-sided Ext characterization	`clusterTilting_iff_nonzero`
Full-object quotient infinity	`allClass_injective`
Complete basic-object quotient count	`basic_class_count`
Actual Coxeter order and exponents	`coxeter_number_eq_two`
	`coxeter_exponents_eq`

The project pins Lean 4.33.0 and Mathlib commit

`db584cd6d46c92f209a44c0f1c829460d327499d`.

First run `lean -version`; then use `lean/verify.sh` for the root build, warning-as-error namespace audit and official full fresh-kernel replay. The accompanying receipts bind exact sources, commands, exit status and original execution data. `Check.lean` enumerates every owned declaration and invokes ordinary `#print axioms` on each, including generated names.

The allowed foundational axioms are only `propext`, `Classical.choice`, and `Quot.sound`.

References

[1] I. Burban, O. Iyama, B. Keller and I. Reiten, *Cluster tilting for one-dimensional hypersurface singularities*, author manuscript, Definition 1.1, printed p. 3. `https://webusers.imj-prg.fr/~bernhard.keller/publ/ClusterSing.pdf`.

[2] A. B. Buan, R. Marsh, M. Reineke, I. Reiten and G. Todorov, *Tilting theory and cluster combinatorics*, arXiv:math/0402054v1 (2004), printed pp. 1–3, 9–11. `https://arxiv.org/pdf/math/0402054v1`.

[3] S. Fomin and N. Reading, *Generalized cluster complexes and Coxeter combinatorics*, arXiv:math/0505085v3 (2006), printed pp. 28–29. `https://arxiv.org/pdf/math/0505085v3`.